LIST OF APPLICABLE CODES AND INDUSTRY STANDARDS

FWEL CONTRACT NO. : **1-14-16601**

CONTRACT NO. : **POGC-760-81-156**

PROJECT : **SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)**

COMPANY : **NATIONAL IRANIAN OIL COMPANY**

SITE : **ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN**

REV.	DATE	DESCRIPTION	ORIG.	PRPD	CHKD	APP'D	CONTRACTOR's APP'D
6	02/02/04	Issued for Company Approval	LG	J.S.Choi	H.C.Shin		
5	21/11/03	Issued for FEED Package	FWEL	G.Neeson	A.Tuckett	A.Tuckett	
4	28/10/03	Issued for FEED Package	FWEL	G.Neeson	A.Tuckett	A.Tuckett	
3	09/10/03	Issued for Client Review	FWEL	G.Neeson	A.Tuckett	A.Tuckett	
2	05/09/03	Issued for FW Int^{l} Review	FWEL	G.Neeson	A.Tuckett	A.Tuckett	
1	24/02/98	Issued by Technip	Technip	P.H.Collet	T.Dupays	P.H.Collet	

TABULATION OF REVISED PAGES

PAGE	REV 4	REV 5	REV.6		
1			X		
7	X				
8	X	X			
10	X	X			
11		X			
12	X				
13	X				
14	X				
17	X				
18	X				
20	X	X			
21	X		X		
22	X				
23	X				
24	X				
25	X				
28	X				

CONTENTS

1. SCOPE

2. INSTRUCTIONS

3. ATTACHMENT

1 SCOPE

This specification provides the list of International Codes and Industry Standards which will govern the design and construction of the ONSHORE FACILITIES for SOUTH PARS GAS FIELD DEVELOPMENT PROJECT (Phases 9 & 10) planned at Assaluyeh in Iran for PARS OIL AND GAS CO.

2 INSTRUCTIONS

2.1 The project facilities shall be designed, engineered, built, installed and commissioned to the latest relevant international codes and standards. Other references stated in these standards shall also apply. These reference documents shall be the latest revision at the date of call for tenders (issued up to 15th September 2002).

2.2 CONTRACTOR shall be responsible for checking consistency of specified documents and informing COMPANY of any interpretation prior to application.

2.3 The list attached concerns only the codes & standards, it is understood that compulsory regulations (local /international) shall be applied.

2.4 The list attached is general and not exhaustive, it shall not restrict the requirement mentioned in the specifications or other documentation.

3 ATTACHMENT

ATTACHMENT 1 - List of applicable codes and industry standards.

ATTACHMENT 1 - List of Applicable Codes and Industry Standards.

TABLE OF CONTENTS

1. PROCESS DESIGN

2. PRESSURE VESSELS - HEAT EXCHANGERS

3. HEATERS - BOILERS

4. AIR COOLERS

5. STORAGE TANKS (NON REFRIGERATED)

6. LPG STORAGE TANKS

7. ROTATING EQUIPMENT

8. OVERHEAD CRANES AND HOISTS

9. PIPING

10. CIVIL - STRUCTURAL STEEL

11. BUILDINGS

12. HVAC

13. ELECTRICAL

14. INSTRUMENTATION - CONTROLS

15. SAFETY / FIRE PROTECTION

16. NOISE CONTROL

17. PAINTING

18. INSULATION

19. FIREPROOFING

20. WELDING / WELDING AND STEEL MATERIAL INSPECTION

21. TELECOM

1 PROCESS DESIGN

API (American Petroleum Institute)

API RP 520 Pt. 1 7th Ed. Jan. 2000	Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries
API RP 521 4th Ed. March 1997	Guide for Pressure-Relieving and Depressurising Systems
API STD 2000 April 1998	Venting Atmospheric and Low Pressure Storage Tanks Non refrigerated and Refrigerated

2 PRESSURE VESSELS - HEAT EXCHANGERS

ASME (American Society of Mechanical Engineers)

ASME SECTION VIII 2001	ASME Boiler & Pressure Vessel Code, Div. 1 and 2 including Addenda (2002)
TEMA 8th Ed. 1999	Standards of the Tubular Exchangers Manufacturers Association
EJMA 7th Ed. 1998	Standards of the Expansion Joints Manufacturers Association, Inc.
NACE MR 0175-2002	Sulphide Stress Cracking Resistant Metallic Materials for Oilfield Equipment.

BSI (British Standard Institute)

PD 5500 : 2000	Specifications for Unfired Fusion Welded Pressure Vessels

Others

UBC: 1997	Uniform Building Code - Volume 2 Division II = Wind Design Division III = Earthquake Design
EEMUA 143 1985	Recommendations for Tube end Welding

API (American Petroleum Institute)

API 662 2nd Ed. April 2002	Plate Heat Exchangers for General Refinery Services

Welding Research Council

WRC March 1979	Bulletin 107

3 HEATERS - BOILERS

API 560 3rd Ed. May 2001	Fired Heater for General Refinery Service
ASME Section 1:1999	Boiler and Pressure Vessel Code
API 530 Oct. 1996	Calculation of Heater Tube Thickness in Petroleum Refineries
ASME B31.1 Dec. 2001	Power Piping
API 535 July 1995	Burners for Fired Heaters in General Refinery Services
ASME B31.3 March 2002	Chemical Plant and Petroleum Refinery Piping

CICIND (International Committee on Industrial Chimneys)

2001	Model Code for Concrete Chimneys Part A - The Shell, with commentaries.
1995	Model Code for Concrete Chimneys Part C - Steel Linings, with commentaries.
1999	Model Code for Steel Chimneys, with commentaries.
1996	Manual on Chimney Protective Coatings.

4 AIR COOLERS

API 661 5th Ed. March 2002	Air Cooled Heat Exchangers for General Refinery Service
EEMUA 143 : 1985	Recommendations for Tube end Welding

5 STORAGE TANKS (NON REFRIGERATED)

API (American Petroleum Institute)

API 620 10th Ed. Feb. 2002	Recommended Rules for Design and Construction of large, welded, low pressure Storage tanks
API 650 10th Ed. Nov. 1998	Welded Steel Tanks for Oil Storage
API 2000 5th Ed. April 1998	Venting Atmospheric and Low-pressure Storage Tanks
API MPM CH2.2A 1995	Method for Measurement and Calibration of Upright Cylindrical Tanks
API MPM CH2.2B 1996	Management of Petroleum Measurement Standards - Tank Calibration - Calibration of Upright Cylindrical Tanks Using the Optical Reference Line Method

UBC (Uniform Building Code)

UBC : 1997	Volume 2 Chapter 16 Division II = Wind Design Division III = Earthquake Design

6 LPG STORAGE TANKS

BS 7777 - 2:1993	Flat Bottomed, Vertical, Cylindrical Storage tanks for Low Temperature Service
BS 8110 - 1:1997	Structural Use of Concrete

7 ROTATING EQUIPMENT

API (American Petroleum Institute)

API 610 8th Ed. 1995	Centrifugal Pumps for Petroleum, Heavy Duty Chemical & Gas Industry Services

API 612 4th Ed. 1995	Special purpose steam turbine for Petroleum, Chemical and Gas Industry Services
API 613 4th Ed. 1995	Special-Purpose Gear Units for Refinery Petroleum, Chemical and Gas Industry Services
API 614 4th Ed. April 1999	Lubrication, Shaft-Sealing, and Control System for Special Purpose Applications
API 616 4th Ed. Aug. 1998	Gas Turbines for Refinery Services
API 617 6th Ed. Feb. 1995	Centrifugal Compressors for Petroleum, Chemical and Gas Service Industries
API 618 4th Ed. June 1995	Reciprocating Compressors for Petroleum, Chemical and Gas Industry Services
API 670 4th Ed. 2000	Vibration, Axial-Position, and Bearing Temperature Monitoring Systems
API 671 3rd Ed. Oct. 1998	Special-Purpose Couplings for Refinery Service
API 672 3rd Ed. Sept. 1996	Packaged Integrally Geared, Centrifugal Air Compressors for Petroleum, Chemical and Gas Industry Services
API 673 *2nd Ed*. Jan 2002	Centrifugal Fans for Petroleum, Chemical and Gas Industry Services
API 674 2nd Ed. June 1995	Positive displacement pumps - Reciprocating
API 675 2nd Ed. Oct. 1994	Positive displacement pumps - Controlled volume
API 676 2nd Ed. Dec. 1994	Positive Displacement Pump - Rotary
API 677 2nd Ed. July 1997	General Purpose Gear Units for Petroleum, Chemical and Gas Industry Services
API 682 1st Ed. Oct. 1994	Shaft Sealing Systems for Centrifugal and Rotary Pumps
NFPA 20 - 1999 Oct. 1999	Standard for the Installation of Stationary Pumps for Fire Protection

ASME (American Society of Mechanical Engineers)

Power Test Code :

~~PTC 9 "Displacement Compressors"~~ - CANCELLED

PTC 10 "Compressors and Exhausters" - 1997

PTC 22 "Gas Turbine Power Plant" - 1997 + errata 1999

ISO (International Organisation for Standardisation)

BS EN ISO 9906 : 2000	Rotodynamic Pumps - Hydraulic Performance Acceptance Tests Grades 1 & 2
ISO 1940-1/2 1st Ed.1986 (1) & 1st Ed.1997 (2)	Mechanical Vibration - Balance Quality Requirements of Rigid Rotors
ISO 5199 March 2002	Technical Specifications for Centrifugal Pumps

Others

AGMA 6011:1998	Association of Gear Manufacturers, Practice for High Speed Helical and Herringbone
ASME B73.1 - 2001	Specification for Horizontal End Suction Centrifugal Pumps

8 OVERHEAD CRANES AND HOISTS

FEM 1.001 3rd Ed. 1998	Rules for the Design of Hoisting Appliances
NFE 52.126 1975	French Norm for Testing Cranes
BS 466 : 1984	Specification for Power Driver overload Travelling Cranes, Semi-Goliath and Goliath Cranes for General Use.
BS 2573	Rules for the Design of Cranes - Classification, Stress Calculations & Design of Mechanisms.

9 PIPING

ANSI/ASME (American National Standards Institute)

ASME B1.1 : 1989	Unified Inch Screw Threads
ASME B1.20.1 : 1983	Pipe Threads, General Purpose
ASME B16.5 : 1996	Steel Pipe Flanges and Flanged Fittings - NPS 1/2" through NPS 24"
ASME B16.20 : 1998	Metallic Gaskets for Pipe Flanges - Ring Joint, Spiral Wound and Jacketed
ASME B16.21 : 1992	Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.25 : 1997	Buttwelding Ends
ASME B16.34 : 1996	Steel Valves - Flanged, Threaded and welding End
ASME B16.47 : 1996	Large Diameter Steel Flanges - NPS 26 through NPS 60
ASME B16.9 : 1993	Factory made Wrought Steel Buttwelding Fittings
ASME B16.11 : 2002	Forged Steel Fittings, Socket Welding and Threaded
ASME B31.1 : 2001	Power Piping
ASME B31.3 : 2002	Process Piping
ASME B31.8 : 2000	Gas Transmission and Distribution Piping System
ASME B36.10M : 1996	Welded and Seamless Wrought Steel Pipe
ASME B36.19M : 1985	Stainless Steel Pipe

API (American Petroleum Institute)

API 594 Nov. 1997	Check Valves : Wafer, Wafer Lug and Double Flanged Type
API 598 7th Ed. Oct 1996	Valve Inspection and Testing

API 600 11th Ed.Oct 2001	Bolted Bonnet Steel Gate Valves for Petroleum and Natural Gas Industries
API 602 7th Ed.Oct 1998	Compact Steel Gate Valves - Flanged, Threaded, Welding and Extended Body Ends
API 5L 42nd Ed.Jan 2000	Specification for Line Pipe
API 6D 22nd Ed.Jan 2002	Petroleum and Natural Gas Industries Pipe Line Transportation Systems - Pipe Line Valves
API 6 FA 3rd Ed.April1999	Specification for Fire Test for Valves
API 15 LR 7th Ed.Aug.2001	Specification for Low Pressure Fiberglass Line Pipe and Fittings

MSS (Manufacturers Standardisation Society of the Valve and Fittings Industry, Inc.)

MSS SP6:2001	Standard Finishes for Contact Faces of Pipe Flanges and connecting ends Flanges of Valves and Fittings
MSS SP25:1998	Standard Marking System for Valves, Fittings, Flanges and Unions
MSS SP44:1996	Steel Pipe Line Flanges
MSS SP69:2002	Pipe Hangers and Supports
MSS SP-75-1988	Specification for High Test Wrought Butt Welding Fittings
MSS SP-80-1997	Bronze Gate, Globe, Angle and Check Valves
MSS SP-95-2000	Swage(D) Nipples and Bull Plugs

Others

BS 5351 : 1986	Specification for Steel Ball Valves for the Petroleum, Petrochemical and Allied Industries
BS 5352 : 1981	Specification for Steel Wedge Gate, Globe and Check Valves 50mm and Smaller for the Petroleum, Petrochemical and Allied Industries

BS 6755-1 1986	Testing of Valves Part 1 : Specification for Production Pressure Testing Requirements
EEJMA STD 7th Ed. 1998	Standard of the Expansion Joint Manufacturers Association, Inc. including Addenda

ASTM (American Society for Testing and Materials)

As described within Project Piping Classes.

10 CIVIL - STRUCTURAL STEEL

10.1 Design Codes

Loading

Uniform Building Code	UBC - 1997
Minimum Design loads for buildings and other structures	ASCE 7-98

Concrete Design

Building Code requirements for structural concrete :

American Concrete Institute (ACI), and more specifically :

A.C.I. 318 M-95:1995 and commentary,

A.C.I. 305R - “Hot weather concreting”,

A.C.I. 301 : 1999

A.C.I. SP 66 : 1994 - ACI Detailing Manual

CIRIA Special publication 31 “Guide to concrete construction in the Gulf Region”.

Steel structure Design

A.I.S.C. 9th Ed. 1989	Manual of American Institute for Steel Construction

Foundation

Code of Practice for foundations of machinery (CP 2012-1) 1974

Welded steel tanks for oil storage API 650 for tank foundations.

Roads design

AASHTO Guide for design of Pavement structures 1993

Overseas Road Note 31 (Fourth Edition) - “A guide to the structural design of bitumen-surfaced roads in tropical and sub-tropical countries”.

Specification for Highway works - Highways agency.

The Stationary Office - 1994

10.2 Construction Standards

American Concrete Institute (A.C.I.)

American Institute of Steel Construction (A.I.S.C.)

A.A.S.H.T.O. standards

10.3 Testing, inspection and acceptance standards

A.S.T.M. standards or equal (B.S., N.F....)

11 BUILDINGS

11.1 Design codes

U.B.C - 1997	Uniform Building Code
SEI / ASCE 7 -98	Minimum design loads for buildings and other structures
A.C.I. 318 / 318R:1995	American Concrete Institute
A.I.S.C.	American Institute for steel construction

11.2 Construction standards

American Concrete Institute (A.C.I.)

American Institute of Steel Construction (A.I.S.C.)

11.3 Testing inspection and acceptance standards

A.S.T.M. Standards or equivalent (B.S., NF….)

11.4 Materials

A.S.T.M. Standards or equal (B.S. - N.F. ....)

11.5 Acoustics

BS EN ISO 140-4:1998	Room-to-Room Sound Attenuation
BS EN ISO 717 -1:1997	Acoustics - Ratings of Sound Insulation in Buildings and of Building Elements. Airborne Sound Insulation.
BS EN ISO 717-2:1997	Acoustics - Rating of sound Insulation in Buildings and of Buildings Elements. Impact sound Insulation.

11.6 Insulation

A.S.T.M. Standards or equivalent (BS, NF,….)

12 HVAC

ASHRAE (American Society of Heating, Refrigeration and Air Conditioning Engineers)

ASHRAE Handbooks and Standards

OCMA (Oil Companies Materials Association)

Oil and Gas Environment

ARI (American Refrigeration Institute)

AFI (American Filtering Institute)

ADC (Air Diffusion Council)

SMACNA (Sheet Metal and Air Conditioning Contractors National Association)

Low Pressure Duct Construction Standards

NFPA 90A - 2002 Standard for the Installation of Air-Conditioning and Ventilating Systems

13 ELECTRICAL

IEC (International Electrical Commission)

and in particular:

IEC 60034 1-11	Rotating Electrical Machines, Parts 1 to 11 inclusive
IEC 62271-100	High Voltage Alternating Current Circuit Breakers
IEC 60072	Dimensions and Output Ratings for Rotating Electrical Machines
IEC 60076	Power Transformers
IEC 60079	Electrical Apparatus for Explosive Atmospheres
IEC 60146	Semiconductor Converters
IEC 60044	Current Transformers
IEC 60186	Voltage Transformers
IEC 60228	Conductors of Insulated Cables
IEC 60255	Electrical Relays
IEC 60269	Low Voltage Fuses
IEC 60282	High Voltage Fuses
IEC 60287	Electric Cables - Calculation of the Current Rating
IEC 60947-4	Low Voltage Switchgear and Control gear
IEC 60298	A.C. Metal-Enclosed Switchgear and Control gear for Rated Voltages above 1kV and Up to and Including 52kV
IEC 60309	Plugs, Socket-outlets and Couplers for Industrial Purpose

IEC 60364	Electrical Installations of Buildings
IEC 60947-3	Low Voltage Switches and Control gear - Pt.3 Switches, Disconnectors, Switch-Disconnectors and Fuse Combination Units
IEC 60439	Low-Voltage Switchgear and Control gear Assemblies
IEC 60470	High Voltage Alternating Current Contactors
IEC 60502	Power Cables with Extruded Insulation and their Accessories for Rated Voltages from 1kV up to 30kV
IEC 60529	Classification of Degrees of Protection Provided by Enclosures for Electrical Equipment
IEC 60726	Dry Type Transformers
IEC 61024	Protection of Structures against Lightning

Others

CENELEC	European Standards and Publications
NFC 13200 1987 + Amd. 1989	High Voltage Electrical Installation

14 INSTRUMENTATION - CONTROLS

ANSI (American National Standards Institute)

B 16 - 34:1996	Valves - Flanged Threaded and Welding End.
B 16 - 104 : 1998	Control Valve Seat Leakage
MC 96 - 1:1982	Temperature Measurement Thermocouples

API (American Petroleum Institute)

Manual of Petroleum Measurement Standards

API RP 521 4th Ed. Mar 1997	Guide for Pressure Relief and Depressurising Systems

API RP 526 5th Ed.June 2002	Flanged Steel Pressure Relief Valves
API RP 527 3rd Ed.July 1991	Seat Tightness of Pressure Relief Valves
API RP 551 1st Ed.May 1993	Process Measurement Instrumentation
API RP 552 1st Ed.Oct.1994	Transmission System
API RP 554 1st Ed.Sept.1995	Process Instrumentation and Control
API RP 555 2nd Ed.Nov2001	Process Analysers
API RP 598 7th Ed.Oct.1996	Valve Inspection and Testing

ASME (American Society of Mechanical Engineers)

ASME, Div. 1 2001 + Add. 2002	Flange Rating when Temperature Exceeds the Limits of Sect. VIII, ap. 2 ANSI B 16-5
ASME, Div. 1 2001 + Add. 2002	Hydraulic Test for Safety Relief Valve Sect. VIII

IEC (International Electrical Commission)

IEC 68.2	Basic Environmental Testing Procedures for Electronic Components and Electronic Equipment
IEC 60079	Electrical Apparatus for Explosive Gas Atmosphere
IEC 60947	Ingress Protection for Dust and Water Jets
IEC 60331	Fire resisting characteristics of electrical cables
IEC 60332	Tests on electric cables under fire conditions (with part 1 and part 3c)
IEC 60529	Classification of Degree of Protection Provided by Enclosures

IEC 60584	Thermocouples
IEC 60751	Industrial Platinum Resistance Thermometer Sensors
IEC 61000	Electromagnetic Compatibility for Industrial-Process Measurement and Control Equipment
IEC 61508	Functional Safety of Electrical/Electronic Safety Related Systems

IEEE (Institute of Electrical and Electronics Engineers)

IEEE C37.90-1 March 2002	Surge Protection Criteria

ISA (Instrument Society of America)

ISA 5.1 : 1984	Instrumentation Symbols and Identification
ISA 75.01.01 April 2002	Flow Equations for Sizing Control Valves
ISA 75.03-1992	Face-to-Face Dimensions for Flanged Globe Style Control Valve Bodies
ISA 18.1 : 1979	Alarm and Sequences

Others

EIA RS 232/422/485 1997	Interface between Data Terminal Equipment and Data Communication Equipment Employing serial Binary Data

15 SAFETY / FIRE PROTECTION

General

IP Code Part 15 2nd Ed. 2002	Area Classification Code for Petroleum Installations

FRENCH REGULATIONS

Arrêté du 4 Septembre 1967

Arrêté du 9 Novembre 1972

Arrêté du 19 Novembre 1975

Circulaire et Instruction du 9 Novembre 1989

Circulaire du 6 Juillet 1990

API (American Petroleum Institute)

API 2021 4th.May 2001	Management of Atmospheric Storage Tank Fires

NFPA (National Fire Protection Association)

NFPA 10 : 1998	Portable Fire Extinguishers
NFPA 11 : 1998	Low-Expansion Foam and Combined Agent Systems
NFPA 11A : 1999	Medium- and High-Expansion Foam Systems
NFPA 11C : 1995	Mobil Foam Apparatus
NFPA 12 : 2000	Carbon Dioxide Extinguishing Systems
NFPA 14 : 2000	Installation of Standpipe and Hose Systems
NFPA 15 : 1996	Water Spray Fixed Systems for Fire Protection
NFPA 16 : 1999	Installation of Deluge Form Water Sprinkler Systems and Foam-water spray systems
NFPA 17 : *2002*	Dry Chemical Extinguishing Systems
NFPA 20 : 1999	Installation of Centrifugal Fire Pumps
NFPA 24 : 1995	Private Fire Service Mains and Their Appurtenances
NFPA 30 : 2000	Flammable and Combustible Liquids Code
NFPA 45 : 2000	Fire Protection for Laboratories using Chemicals
NFPA 58 : 2001	Storage and Handling of Liquefied Petroleum Gases
NFPA 59 : 2001	Standard for the Storage and handling of LPG at Utility Gas
NFPA 72 Aug. 2002	The Installation, Maintenance, and Use of Protective Signaling Systems
NFPA 780	Lightening Protection Code
NFPA 101 : 2000	Safety to Life from Fire in Buildings and Structures
NFPA 86 : 1999	Ovens and Furnaces

NFPA 86C : 1999 Industrial Furnaces using a special processing Atmosphere

16 NOISE CONTROL

EEMUA (Engineering Equipment and Material Users Association)

Pub. 140 Noise Procedure Specification (formally OCMA Spec. NWG1, Rev.2, 1980)

ISO (International Organisation for Standardisation)

ISO Standards Handbook 35 Acoustics-Inc : Basic Standards, Methods of Noise Measurement, Audiometry & Human exposure to noise

IEC (International Electrical Commission)

IEC 61260 Aug. 1995 Specification for Octave-Band and Fractional-Octave-Band-Analog and Digital Filters

IEC 60651 Oct. 2001 Sound Level Meters

17 PAINTING

ISO (International Organisation for Standardisation)

ISO 8501Preparation of Steel Substrates before Application of Paints and Related Products -- Visual Assessment of Surface Cleanliness -- Part 1: Rust Grades and Preparation Grades of Uncoated Steel Substrates and of Steel Substrated after Overall Removal of Previous Coatings.

SSPC (Steel Structures Painting Council)

SSPC AB1 : 1991 Abrasive Specification No.1, Mineral and Slag Abrasives

SSPC-PA 2 : 1996 Measurement of Dry Film Thickness with Magnetic gauges

SSPC-PA 3 : 1982 A guide to Safety in Paint Application

SSPC VIS1 : 1989 Visual Standard for Abrasive Blast Cleaned Steel

SSPC VIS3 : 1993 Visual Standard for Power and Hand Tool Cleaned Steel

SP1 : 1982	Solvent Cleaning
SP3:1982	Power Tool Cleaning
SP5:2000	White Metal Blast Cleaning
SP6:2000	Commercial Blast Cleaning
SP7 : 2000	Joint Surface Preparation Standard : Brush-off Blast Cleaning
SP8:1982	Pickling
SP10:2000	Near-White Blast Cleaning
SP11:1987	Power Tool Cleaning to Bare Metal

ASTM (American Society for Testing and Materials)

ASTM D520 : 2000	Standard Specification for Zinc Dust Pigment
ASTM D3359 : 2002	Standard Test Methods for Measuring Adhesion by Tape Test
ASTM D4285 : 1983	Standard Test Method for Indicating Oil or Water in Compressed Air
ASTM D4417 : 1993	Field Measurement of Surface Profile of Blast Cleaned Steel
ASTM D4541 : 1995	Standard Test Method for Pull-off Strength of Coatings Using Portable Adhesion
ASTM D4752 : 1998	Standard Test Method for Measuring MK resistance of Ethyl Silicate (Inorganic) Zinc Rich Primers by Solvent Rub
ASTM E337 : 2002	Standard Test Method for Measuring Humidity with a Psychrometer - Measurement of Wet and Dry Bulb Temperatures

BSI (British Standards Institute)

BS 1710 : 1984	Specifications for Identification of Pipelines and Services
BS 7079 : 1990	Preparation of Steel Substrates before Application of Paints and Related Products

BS EN ISO 7783-1:2000 Methods of Tests for Paints

NACE (National Association of Corrosion Engineers)

NACE MR 0175:1999	Sulphide Stress Cracking Resistant Metallic Materials for Oilfield Equipment.
NACE RP 0178 : 1989	Fabrication Details, Surface Finish Requirements and Proper Design Considerations for Tanks and Vessels to be Lined for Immersion Service
NACE RP 0188 : 1999	Discontinuity (Holiday) Testing of Protective Coatings
NACE RP 0288 : 1994	Inspection of Linings on Steel and Concrete
NP 6 H 189	A State-of-the-Art Report of Protective Coatings for Carbon Steel and Austenitic Stainless Steel Surfaces Under Thermal Insulation and Cementitious Fireproofing
TM-O1	Visual Standard for Surfaces of New Steel Air blast Cleaned with Sand Abrasives

Others

NFT 30 124 : 1991	Determination of Dry Film Thickness Non Destructive Instrumental Methods
DIN 55928-9 : 1991	Corrosion Protection of Steel Structures by the Application of Organic or Metallic Coatings

18 INSULATION

ASTM (American Society for Testing and Materials)

ASTM A167:1999	Stainless Steel and Heat Resisting Chromium - Nickel Steel Plate, Sheet and Strip
ASTM A463 : 2001	Steel Sheet, Cold-Rolled, Aluminum-coated, Type 1 and Type 2
ASTM C165 : 2000	Measuring Compressive Properties of Thermal Insulations
ASTM C167 : 1998	Fiberglass Material Density
ASTM C240 : 1997	Foamglas Density
ASTM C450 : 1999	Prefabrication and Field Fabrication of Thermal Insulating Fitting Covers, Adjunct No. 12-304500-00
ASTM C461 : 1981	Mastics and Coatings Used with Thermal Insulation (R 1990)

ASTM C518 : 1998	Steady-Sate Heat Flux Measurements and Thermal Transmission Properties by Means of The Heat Flow Meter Apparatus
ASTM C547 : 2000	Mineral Fiber Performed Pipe Insulation
ASTM C552 : 2000	Cellular Glass Thermal Insulation
ASTM C578 : 2001	Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM C591 : 2001	Unfaced Preformed Rigid Cellular Polyurethane Thermal Insulation
ASTM C592 : 2000	Mineral Fiber Blanket Insulation & Blanket Type Pipe Insulation (metal mesh covered)
ASTM C612 : 2000	Mineral Fiber Block and Board Thermal Insulation
ASTM C680 : 1989	Determination of Heat Gain or Loss and the Surface Temperature of Insulated pipe and Equipment Systems by the Use of a Computer Program
ASTM C795 : 1992	Wicking Type Thermal Insulation for use over austenitic stainless steel
ASTM C871 : 1995	Thermal Insulation Materials for Leachable Chloride, Fluoride, Silicate, and Sodium Ions
ASTM D93 : 2000	Flash Point by Pensky-Martens Closed Tester
ASTM D1621 : 2000	Compressive Properties of Rigid Cellular Plastics (R 1979)
ASTM D1622 : 1998	Apparent Density of Rigid Cellular Plastics
ASTM D1623 : 1978	Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics
ASTM D2856 : 1994	Open Cell Content of Rigid Cellular Plastics by the Air Pycnometer
ASTM E84 : 2001	Surface Burning Characteristics of Building Materials
ASTM E90 : 1999	Measurement of Airborne Sound Transmission Loss in Building Partitions
ASTM E96 : 2000	Water Vapour Transmission of Materials

BSI (British Standard Institute)

BS 4735 : 1974	Laboratory Method of Test for Assessment of the Horizontal Burning Characteristics of Specimens no longer than 150mm x 50mm x 13mm of Cellular Plastics and Cellular Rubber Materials when subjected to a Small Flame
BS 5970 : 2001	Thermal Insulation of Pipe Work and Equipment (in the temperature range of -100°C to +870°C)

19 FIREPROOFING

UL 1709 : 1994	Fire Tests on Building Materials and Structures (Underwriter Laboratories)

20 WELDING / WELDING AND STEEL MATERIAL INSPECTIONS

ASME (American Society of Mechanical Engineers) Codes

Section II	Part C : Welding Rods, Electrodes and Filler Metals
Section V	Non-destructive Examination
Section VIII	Unfired Pressure Vessels, Divisions 1 and 2
Section IX	Welding Qualification

European Norms

EN 10204 1991 + Add. 1 1995	Metallic Products - Types of Inspection Documents

British Standards

BS EN 287-1 : 1992	Approval Testing of Welders
BS EN 288-9 : 1999	Welding Procedure for Metallic Materials
BS EN 462 1-5 1994 - 1997	Non Destructive Testing IQI's
BS EN 473 : 2000	Qualification and Certification of Non-destructive Personnel

BS EN 571-1 : 1997	Non Destructive Testing. Penetrant Testing - General Principles
BS EN 584 Pt. 1 - 1995, Pt. 2 - 1997	Non-destructive testing - Industrial Radiographic Film
BS EN 1435 : 1997	Non Destructive Examination of Welds. Radiographic Examination of Welded Joints
BS EN 1714 : 1998	Ultrasonic testing of welded joints
BS 3923-2 : 1972	Methods for Ultra Sonic Examination of Welds
BS 4871 Pt.3 : 1985	Welder Qualification - Tube to Tubesheet Joints
BS EN ISO 9934-1 2001	Non Destructive Testing - Magnetic Particle Testing - General Principle
BS EN 10160 : 1999	Ultra Sonic Testing of Steel Flat Products (greater than 6mm

ASTM (American Society for Testing and Material(s) Standards

E92 : 1982	Test Method for Vickers Hardness of Metallic Materials
E94 : 2000	Guide for Radiographic Testing
E114 : 1995	Ultrasonic Pulse-Echo Straight Beam Examination by the Contact Method
E125 : 1963	Magnetic Particle Indications on Ferrous Castings
E165 : 2002	Liquid Penetrant Inspection Examination
E709 : 2001	Magnetic Particle Examination
A578 : 1996	Specification for Straight-Beam Ultrasonic Examination of Plain and Clad Steel Plates for Special Applications
A833 : 1984	Indentation Hardness of Metallic Materials by Comparison Hardness Testers
E110 : 1982	Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers
E140 : 2002	Standard Hardness Conversion Tables of Metals

AWS (American Welding Society)

AWS A 4-2:1997	Standard Procedure for Calibrating Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic and Duplex Austenitic-Ferritic Stainless Steel Weld Metal.
AWS A5.1 : 1991	Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding
AWS A5.4 : 1992	Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding
AWS A5.5 : 1996	Specification for Low Alloy Steel Covered Arc Welding Electrodes
AWS A5.9 : 1993	Specification for Bare Stainless Steel Welding Electrodes and Rods
AWS A5.18 : 2001	Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS A5.28 : 1996	Specification for Low Alloy Steel Electrodes and Rods for Gas Shielded Arc Welding

ASNT (American Society of Non-destructive Testing)

ASNT SNT TC 1A 2001	Recommended Practice for Non-Destructive Testing Personnel Qualification and Certification.

21 TELECOM

Where there are no equivalent Iranian standards, the appropriate International and European standards shall be applied to the Telecommunications system Design. These shall include, but not be limited to, all applicable sections of :-

- ITU Recommendations, including ITU-R and ITU-T
- EIA, IEE, IEEE, ISO and IEC Standards
- DTI MPT Specifications
- IMO Regulations
- ICAO Regulations
- CENELEC Standards